

Responsible AI for Climate-Dependent Agricultural Systems

Data, Inclusivity and Trustworthy Decision Support

Building agricultural AI that is useful, fair, transparent and grounded in local realities.

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About me

My Background

Data Scientist and Geospatial Professional, with a background in GEGIS, currently pursuing an MSc. in Data Science and Analytics

What I work on

I work with agricultural data, GIS, remote sensing, AI (machine learning), crop modelling support and decision-support tools.

I also have experience in developing AI-ready data pipelines

My current focus

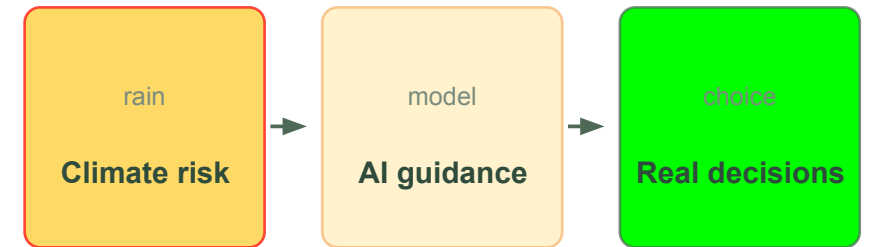
At IITA-CGIAR, my work supports agronomic data standardization, reproducible data workflows, fertilizer recommendation modelling, and the translation of complex agricultural, soil, climate and geospatial data into practical insights.

Why this topic matters to me

Responsible AI matters because the outputs we produce are not just model results; they can influence real decisions made by farmers, researchers, extension teams and policy partners.

Why this conversation matters

- Agriculture is increasingly shaped by delayed rains, droughts, floods, heat stress and shifting seasons.
- AI can translate complex data into practical guidance, but poor recommendations can affect real livelihoods.
- Responsible AI asks whether the tool is accurate, useful, fair, explainable and grounded in local realities.
- The goal is not smarter models alone. The goal is better decisions.



Responsible AI in agriculture is a human-centred practice: it starts with the people who use the advice.

Start with the farmer, not the algorithm

The real question is often simple:

When should I plant?

What should I apply?

What risk am I taking?

- AI systems should support farmers, extension officers, agronomists and project teams, not remove judgement from the process.
- A responsible tool respects context: soils, crops, prices, climate, labour, access to inputs and risk tolerance.

Local advisory

For maize in the long rains season:

Target yield: X t/ha

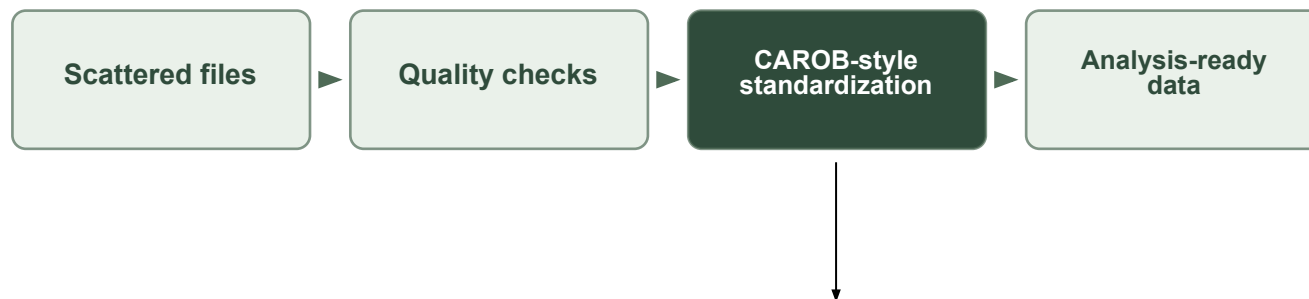
Apply: j kg N + k kg P

Translate into familiar inputs:
DAP / MEA Mazao / Mavuno
(bags or kilograms)

Simple channels: extension, SMS, USSD,
radio, farmer apps

Good AI begins with good data

- Agricultural AI depends on trials, surveys, soil records, climate data, yield observations and geospatial layers.
- The data is often messy: missing values, inconsistent units, wrong coordinates, unclear metadata and uneven coverage.
- Curation, standardization, quality control, documentation and FAIR workflows matter before modelling begins.



That cleans and transforms agricultural surveys into a standard format, and aggregates larger databases that can be used in further analysis.

dataset_id	country	site
doi_10.18167_DVN1_DLTQWR	Nigeria	University Farm, Ibadan
doi_10.18167_DVN1_DLTQWR	Nigeria	University Farm, Ibadan
doi_10.18167_DVN1_DLTQWR	Nigeria	University Farm, Ibadan
doi_10.18167_DVN1_DLTQWR	Nigeria	FOC University of Nigeria, Nsukka
doi_10.18167_DVN1_DLTQWR	Nigeria	FOC University of Nigeria, Nsukka
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doi_10.18167_DVN1_DLTQWR	Nigeria	FOC University of Nigeria, Nsukka
doi_10.18167_DVN1_DLTQWR	Ethiopia	Adigudem
doi_10.18167_DVN1_DLTQWR	Ethiopia	Adigudem

CAROB helps make data comparable, reusable and transparent.

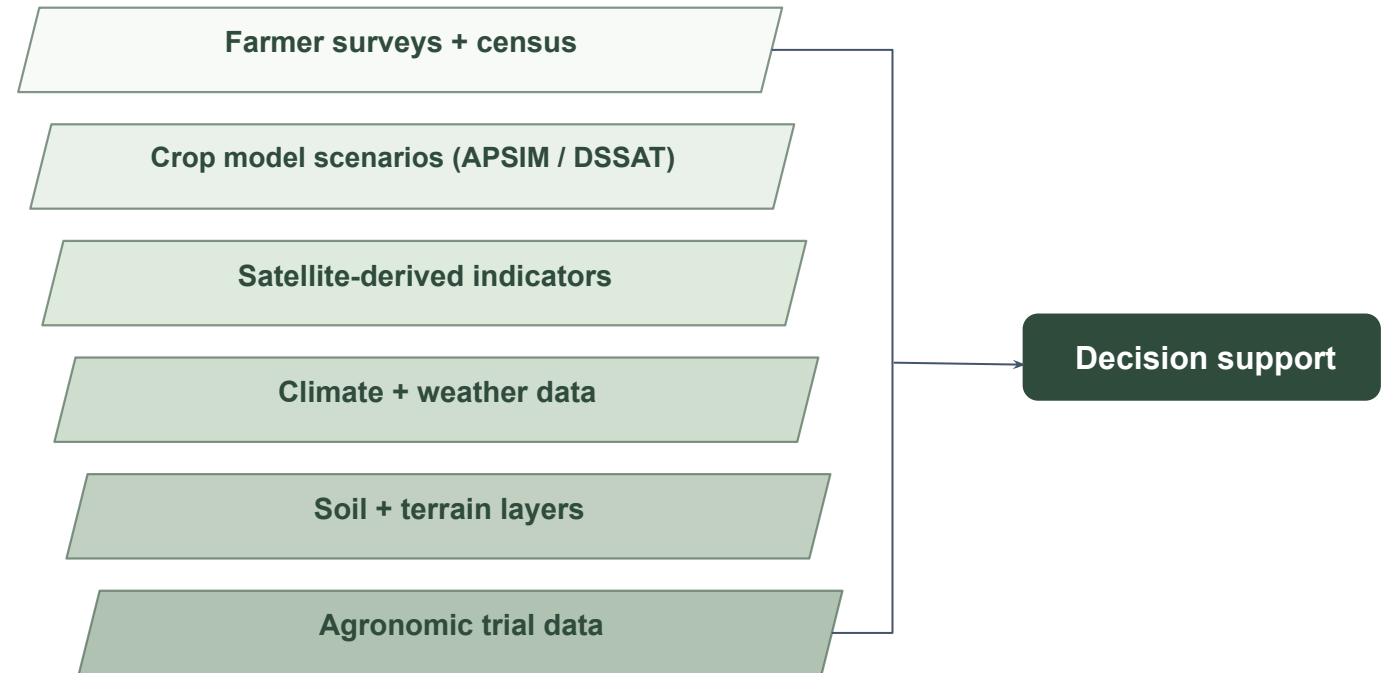
CAROB is an open, reproducible workflow that cleans, standardizes, documents, and combines agricultural research data from experiments and surveys into reusable datasets for further analysis and decision support.

Learn more about CAROB: <https://carob-data.org/>

My lens: turning raw data into decision-ready evidence

This is where responsible AI becomes practical: we make the data usable, explainable and fit for decision support.

- **CAROB-style standardization:** clean, harmonize and document agronomic and survey datasets.
- **Integrated agricultural intelligence:** combine agronomic, soil, climate and geospatial data for modelling and fertilizer recommendation insights.
- **Filling data gaps responsibly:** use satellite-derived data, crop model scenarios, census data and farmer surveys where field data is limited.



Climate uncertainty changes the rules

- A model trained on historical seasons may struggle when rainfall patterns, temperatures and extremes shift.
- Responsible systems should avoid false certainty and communicate confidence, risk and assumptions clearly.
- Scenario analysis, remote sensing, crop models and climate data help explore possibilities rather than one perfect answer.

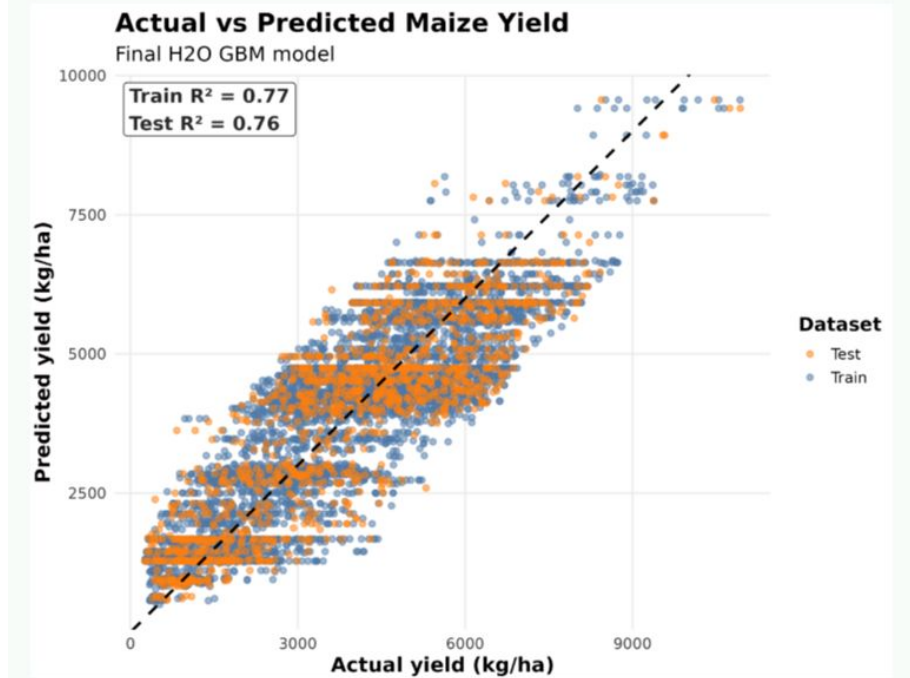
Below normal

Normal

Above normal

Recommendations can be presented as scenario-aware guidance, not absolute predictions.

Gradient Boosting Machine (GBM) was the best performing model with a test R^2 of 76%

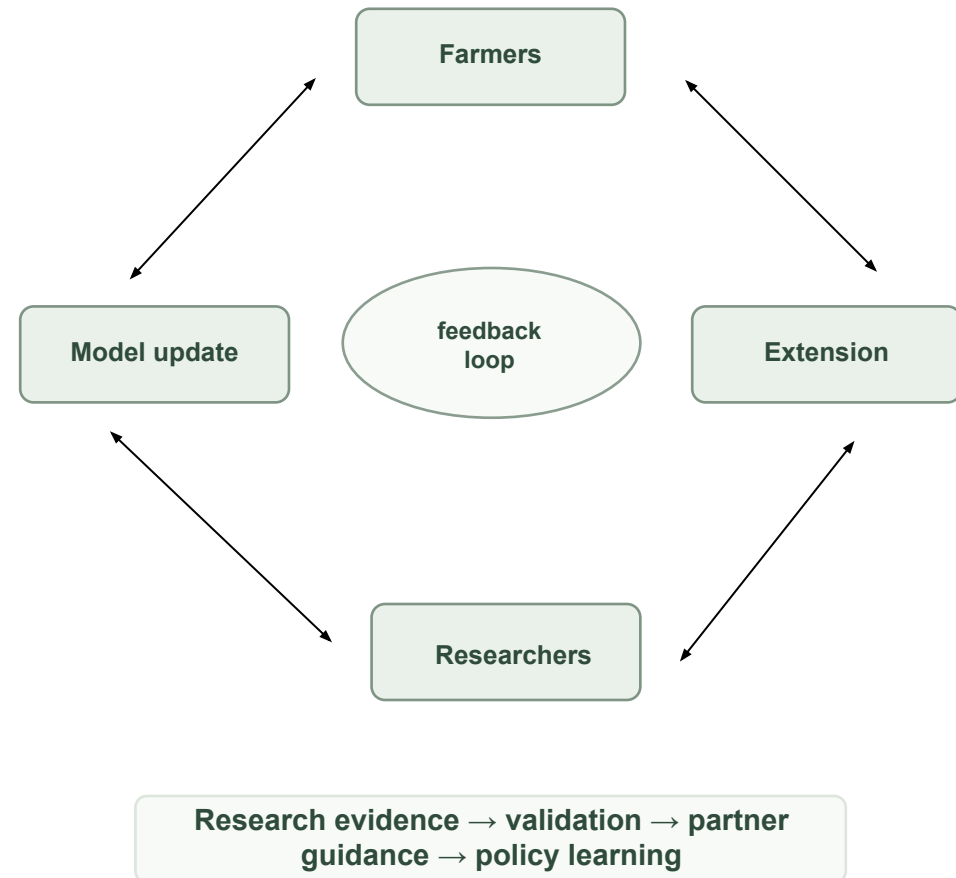


For technical users: report validation results, accuracy, error and data coverage.

For non-technical users: translate into high / medium / low confidence.

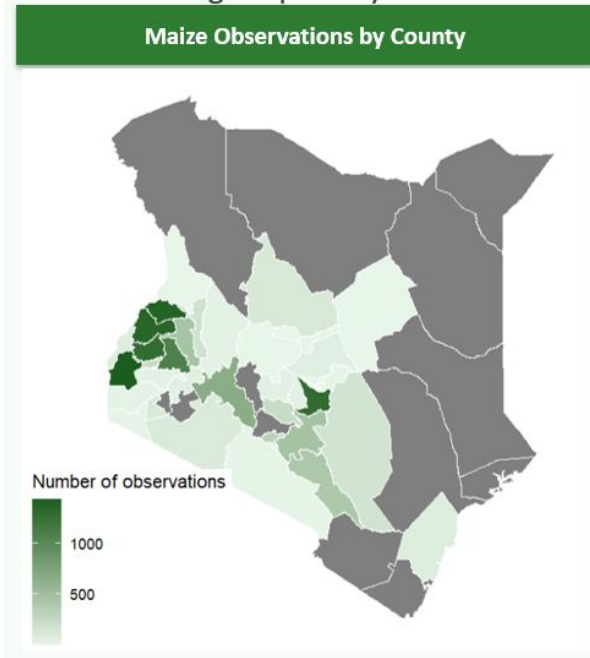
Building trust over time

- Trust grows when outputs are tested across seasons, regions, soil types, management practices and farmer groups.
- **Feedback loops matter:** farmers, extension officers, researchers and partners should help improve the system over time.
- Documentation, version control, transparent workflows and technical reporting make models easier to review, reproduce and improve.



Inclusivity: who is represented?

- AI reflects the data used to build it. If some regions, crops, soils or farmer groups are missing, the tool may serve them poorly.
- Responsible design asks whose conditions are captured, whose constraints are understood, and who can actually use the final advice.
- Inclusivity means working with local partners, extension teams and farmers, not only with datasets and algorithms.



Representation checks

Regions

Soil types

Farm sizes

Gender/youth

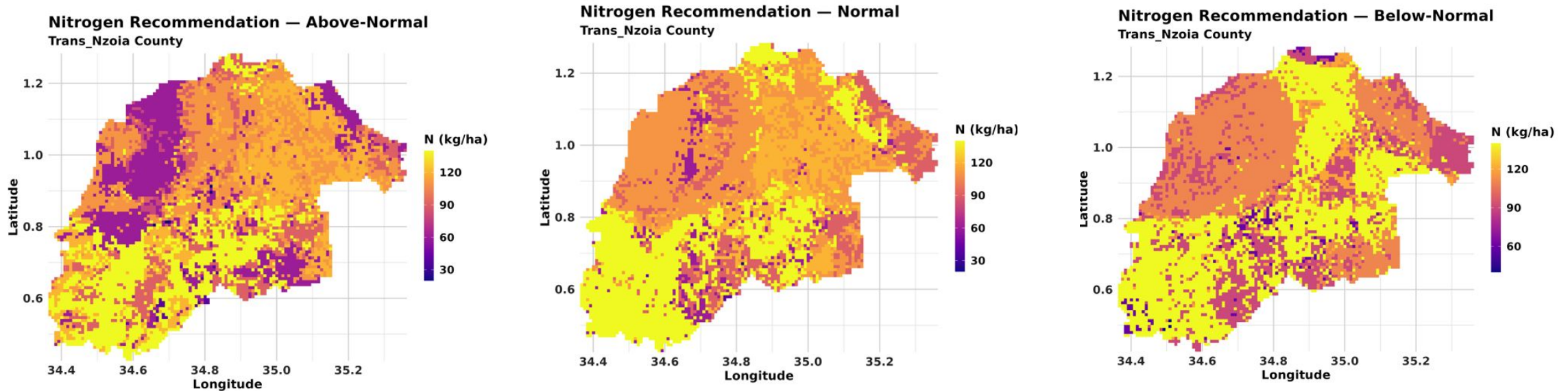
Input access

PWDs and
Vulnerable groups

Fairness is not a final model test only. It starts with data collection, curation, validation and communication.

Trustworthy decision support

- Trustworthy tools explain the basis of recommendations: data sources, assumptions, model logic and limitations.
- Recommendations should be locally validated, understandable and connected to agronomic knowledge and field realities.
- The best systems help people compare options, understand trade-offs and make informed decisions under uncertainty.



A map is useful when paired with plain-language advice:
“for this location and season, apply this range of fertilizer.”

Closing thought

Responsible AI for climate-dependent agriculture is a practice of care:

care for the data, care for the people represented by the data, and care for how recommendations are used.

Reliable data

quality, metadata, standards

Inclusive design

local context and representation

Trustworthy support

uncertainty, validation and feedback

Better data. Better tools. Better decisions.

Thank you!